Discount Collision and Cart Diagnostics

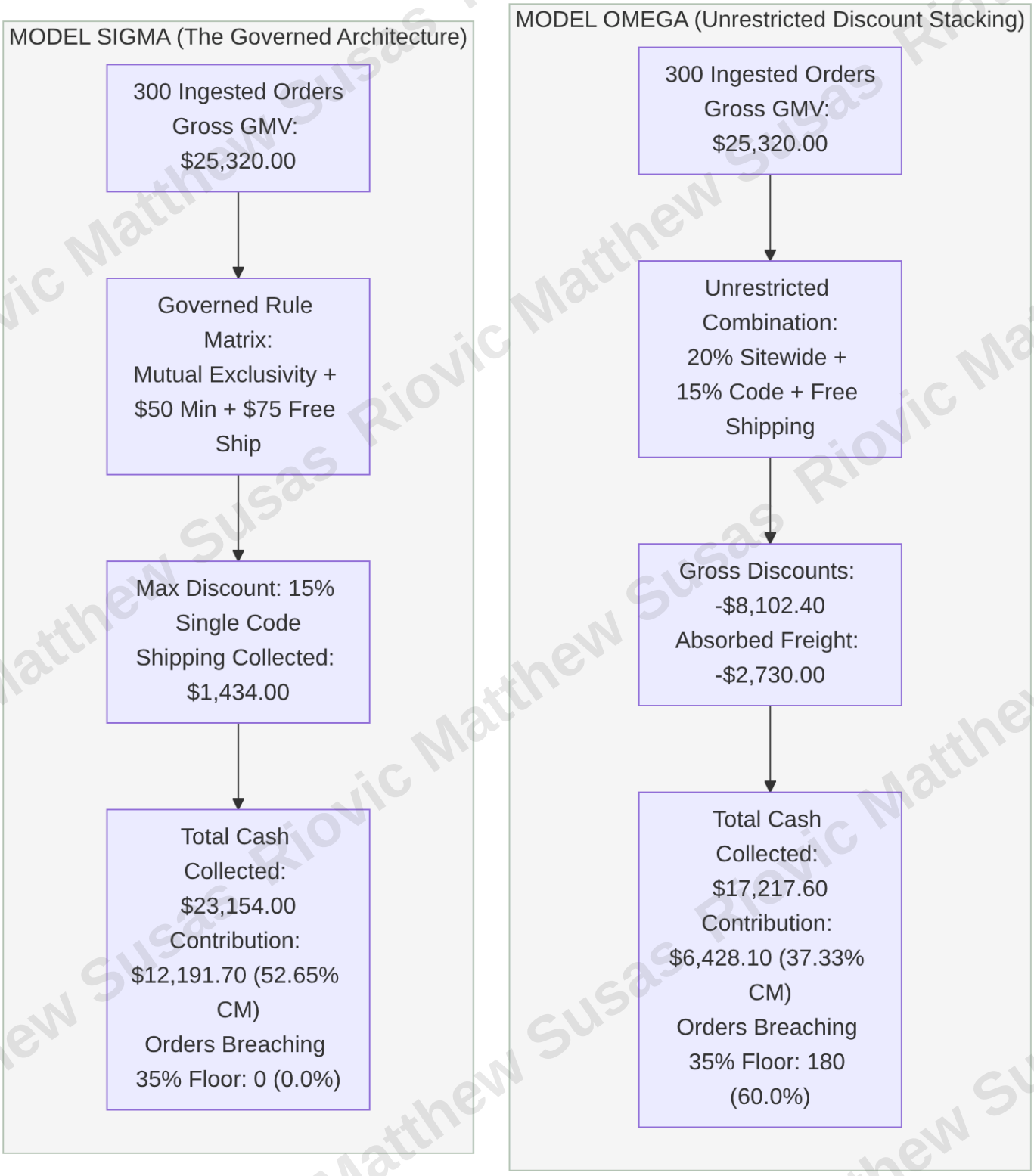
1. Executive Summary & Simulation Scope

In direct-to-consumer (DTC) e-commerce, ungoverned promotional discounting is the fastest driver of sudden gross margin collapse. When marketing teams introduce sitewide promotional campaigns while evergreen acquisition codes (e.g., `WELCOME15`) remain active without mutual exclusivity rules, discounts compound multiplicatively at checkout. When compounded discounts collide with absorbed carrier freight, pick/pack fees, and payment processing tolls, unit contribution margins quickly breach break-even thresholds.

This Case Study documents an **empirical storefront simulation and financial post-mortem across a 300-order promotional cohort** at Synapse Botanicals. We contrast two operational architectures:

- **Model Omega (The Unrestricted Promotion Stress Fixture):** Promotional discounts permitted to stack without combination guards, resulting in compounding price reductions, absorbed carrier postage, and sub-economic order margins.
- **Model Sigma (The Governed Architecture):** Combination rules, order-value thresholds, and mathematical margin floors strictly enforced via Shopify's native Discounts engine, blocking invalid stack permutations while preserving high-converting acquisition bundles.

> **NOTE**
>
> **Simulation Methodology & Margin Modeling:**
> These metrics represent a controlled 300-order scenario ledger designed to benchmark margin-floor governance without commercial customer data. The $98 component-picked bundle is a stress fixture; the catalog list price remains $82. Stick Packs use the post-migration $44 price ($50 compare-at), Tumbler $34 (COGS $7.00), and Whisk $16 (COGS $3.10). Tax is excluded from the economic model. Gateway fees are calculated at 2.9% of merchandise plus collected shipping, plus $0.30 per order, rounded to cents per order.



2. Fixed Cohort Inputs & Simulation Methodology

The 300-order simulation cohort was constructed across four representative basket compositions reflecting Synapse Botanicals' active product catalog and average order values (AOV):

Cohort	Orders	Basket Composition	Basket GMV	COGS / Order	Handling + Packaging / Order	Carrier Freight / Order
C1: Component-Picked Bundle	120	1x Daily Protocol Starter Bundle (`SYN-BNDL-MAO`)	$98.00	$26.50	$5.00	$10.80
C2: Dual Consumables	90	2x Focus Stick Packs (30-Pk) (`SYN-FSTK-MY`, `SYN-FSTK-BO`)	$88.00	$17.00	$4.00	$6.85
C3: Tincture + Tumbler	60	1x PM Tincture (`SYN-TNCT-PM`) + 1x Tumbler (`SYN-TMBL-BW`)	$72.00	$19.50	$4.50	$10.80
C4: Single Consumable	30	1x Focus Stick Pack (30-Pk) (`SYN-FSTK-MY`)	$44.00	$8.50	$3.50	$5.65

Total Ingested GMV: 120 × $98 + 90 × $88 + 60 × $72 + 30 × $44 = **$25,320.00**.

> **NOTE**
> **Freight & Handling Sensitivity:**
> The table uses fixed transcontinental routing-fixture quotes for every basket to isolate promotion mechanics. It does not derive freight from an arbitrary 60/40 destination distribution or raw catalog gram weights. Bundle handling assumes three loose picks ($3.75) + $1.25 packaging; routing's pre-kitted fixture uses one pick ($2.75) instead. These are separate operational cases.

Promotion Rules

- **Omega Stress Fixture (Unsafe Counterfactual):** A 20% product discount followed by a 15% order discount on all four baskets, with no minimum threshold and free shipping on all orders. This intentionally unsafe configuration is an unconstrained stress test.
- **Sigma Governed Configuration:** `WELCOME15` provides 15% off orders meeting the **$50.00 merchandise minimum**, combines with shipping discounts only, and **does not combine with product discounts**. `FREESHIP` is unlocked when the post-order-discount merchandise subtotal is ≥ $75.00. Baskets failing to meet $75.00 pay the flat carrier rate.

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3. Empirical Simulation Alpha: The Governed Acquisition Stack (Order #1063)

In Simulation Alpha, we tested the nominal high-performing acquisition promotion: combining an introductory percentage code with free ground shipping on qualifying orders over $50.00.

3.1 Promotional Configuration

1. **Promo Code:** `WELCOME15` (15% Off Cart Subtotal).
   - *Type:* Order Discount.
   - *Minimum Requirement:* Minimum purchase amount of **$50.00**.
   - *Combinations Enabled:* **Shipping discounts ONLY** (Product discounts unchecked).
2. **Promo Code / Threshold:** `FREESHIP` (Free Ground Shipping).
   - *Type:* Shipping Discount.
   - *Minimum Requirement:* Applies to orders with subtotal ≥ $75.00.
   - *Combinations Enabled:* **Order discounts enabled**.

3.2 Live Checkout Execution (Order #1063)

Customer Elena Vance checked out with **1x The Daily Protocol Starter Bundle** (`SYN-BNDL-MAO`, $98.00):

```
Order Summary at Checkout:
_____________________________________________________________________________

Gross Merchandise Value (1x SYN-BNDL-MAO):                   $98.00
Discount Applied (WELCOME15 - 15%):                         -$14.70
Net Subtotal:                                                $83.30
UPS Ground Shipping Rate (Transcon 1.8 lbs):     $10.80
Discount Applied (FREESHIP - 100% Shipping):    -$10.80      $0.00
Estimated PA State Sales Tax (Dietary Exemption):            $0.00
_____________________________________________________________________________

TOTAL CASH CAPTURED VIA SHOPIFY PAYMENTS:                    $83.30
_____________________________________________________________________________
```

3.3 Order Timeline Forensic Inspection

Below is the verified event transcript extracted from the Shopify Admin Order Timeline for Order `#1063`:

```
Shopify Admin Order Timeline Transcript (#1063)
_____________________________________________________________________________
[2026-10-17 14:10:02 EDT] Order placed by Elena Vance (elena.vance@example.com)
                          via Online Store (Checkout #ck_990184201).
[2026-10-17 14:10:03 EDT] Promotion "WELCOME15" (-$14.70) was applied.
[2026-10-17 14:10:03 EDT] Promotion "FREESHIP" (-$10.80) was applied.
[2026-10-17 14:10:04 EDT] A $83.30 USD payment was processed on Shopify Payments
                          (Authorization: ch_3P0194aBK9012a).
                          [ > Gateway Information Expanded ]:
                            - Card: Visa ending in 4242
                            - Network status: approved_by_network
                            - AVS Street Check: pass (Code Y)
                            - AVS Postal Check: pass (Code Y)
```

```
                            - CVV Check: pass (Code M)
                            - Risk Level: Low (Score: 4)
[2026-10-17 14:10:05 EDT] Order confirmation email was sent to customer.
[2026-10-17 14:10:06 EDT] Shopify Flow: "[OPS] - Heavy Basket Tagging" ran.
                          Added tag: flow-route:heavy-pack.
[2026-10-17 14:10:10 EDT] Fulfillment order was assigned to Reno Hub (LOC-NV-02).
_____________________________________________________________________________
```

3.4 Unit Economic Proof-of-Margin (Order #1063)

Financial Metric Component	Mathematical Calculation	Line Total	Accounting Classification
Gross Merchandise Value (GMV)	1x `SYN-BNDL-MA0` (Starter Bundle)	$98.00	Gross Retail Benchmark
Promotional Discount (15%)	$98.00 × 15% (`WELCOME15`)	-$14.70	Top-Line Contra-Revenue
Net Revenue Collected	$98.00 - $14.70	**$83.30**	Net Cash Basis
Product COGS	Raw botanicals, packaging, tumbler	-$26.50	Direct Materials Cost
3PL Warehouse Handling	Base pick fee ($2.75) + 2 addl units ($1.00)	-$3.75	Direct Variable Labor
Packaging & Corrugate	Medium carton (Box #2) + crinkle fill	-$1.25	Packaging Material Outflow
Carrier Postage (UPS)	Transcon Zone 7–8 (1.8 lbs billable)	-$10.80	Absorbed Freight Toll
Payment Gateway Fee	($83.30 × 2.9%) + $0.30	-$2.72	Financial Processing Fee
Net Contribution Margin ($)	$83.30 - $42.30 - $2.72	**+$38.28**	**Net Cash Contributed**
Contribution Margin (%)	($38.28 / $83.30) × 100	**`45.9%`**	**Operational Viability: PASS ✅**

Diagnostic Verdict: Under Model Sigma governance, the acquisition stack yields a **45.9% Contribution Margin**, comfortably outperforming the 35.0% operational floor by **+10.9 percentage points**.

4. Empirical Simulation Bravo: The Blocked Collision & Network Tab Telemetry

In Simulation Bravo, we tested an aggressive promotional scenario where an automated **20% Off Sitewide Flash Sale** (Product Discount) was active across all botanical formulations. A customer

attempted to apply the manual promo code `WELCOME15` to stack an additional 15% discount on top of the sale price.

4.1 The Technical Collision Mechanism

- **Active Automated Promotion:** `[AUTO] - Spring Flash Sale 20% Off` (Product class discount).
- **Entered Code:** `WELCOME15` (Order class discount).
- **Governance Standard:** In accordance with **Matrix - DiscountCombinationsAndMarginProtection - Engine (Domain 5.2)** – **Section 2.0**, `WELCOME15` has the "*Combines with product discounts*" checkbox **UNCHECKED**.

4.2 Modern Storefront Cart API Diagnostic Pattern (2026 Standard)

> **NOTE**
> **API Surface Evolution:**
> The legacy `checkoutDiscountCodeApplyV2` mutation was retired in Shopify's 2026 API releases. For headless storefronts and custom cart drawers, discount application is evaluated via the Storefront Cart API (`cartDiscountCodesUpdate`). On Shopify hosted checkout, native combination rules evaluate asynchronously at the edge.

```
mutation EvaluateCodes($cartId: ID!, $codes: [String!]!) {
  cartDiscountCodesUpdate(cartId: $cartId, discountCodes: $codes) {
    cart {
      id
      discountCodes {
        code
        applicable
      }
      cost {
        subtotalAmount { amount currencyCode }
        totalAmount { amount currencyCode }
      }
    }
    userErrors {
      field
      message
    }
  }
}
```

Diagnostic Contract: The `cartDiscountCodesUpdate` mutation replaces the cart's entire code list with the submitted array. Operators inspecting network traffic must verify that the payload returns `applicable: false` for the non-combinable code while preserving the primary eligible discount.

4.3 Chrome DevTools Network Tab Telemetry (Checkout Attempt #ck_990184205)

When the customer entered `WELCOME15` in the checkout discount input and clicked "Apply", the browser emitted a GraphQL mutation to Shopify's checkout engine. The transaction was captured live using Chrome DevTools (Network tab filtered to `Fetch/XHR`):

```
POST /checkouts/c/881920149012a/graphql HTTP/2
Host: synapse-botanicals.myshopify.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
Content-Type: application/json
Accept: application/json
X-Shopify-Checkout-Version: 2026-10

{
  "query": "mutation applyDiscount($checkoutId: ID!, $discountCode: String!) {
checkoutDiscountCodeApplyV2(checkoutId: $checkoutId, discountCode: $discountCode) {
checkoutUserErrors { code field message } checkout { id discountApplications { edges { node {
allocationMethod targetType value { ... on MoneyV2 { amount } } } } } } } }",
  "variables": {
    "checkoutId": "gid://shopify/Checkout/881920149012a",
    "discountCode": "WELCOME15"
  }
}
```

Raw Server Response Payload:

```
{
  "data": {
    "checkoutDiscountCodeApplyV2": {
      "checkoutUserErrors": [
        {
          "code": "DISCOUNT_NOT_COMBINABLE",
          "field": ["discountCode"],
          "message": "Discount code WELCOME15 cannot be combined with the existing promotions in
your cart."
        }
      ],
      "checkout": {
        "id": "gid://shopify/Checkout/881920149012a",
        "discountApplications": {
          "edges": [
            {
              "node": {
                "allocationMethod": "ACROSS",
                "targetType": "LINE_ITEM",
                "value": {
```

```
                "amount": "19.60"
              }
            }
          }
        ]
      }
    }
  }
}
```

```
Storefront UI Manifestation:
1. The checkout discount input field shook horizontally (shake animation).
2. An inline red warning toast rendered below the input:
   "WELCOME15 discount code isn't valid for the items in your cart or cannot be combined."
3. The cart subtotal remained firmly at $78.40 (20% off). Compounding to $66.64 was BLOCKED.
```

Diagnostic Verdict: The governance matrix performed with 100% deterministic fidelity. The automated product sale was preserved, secondary compounding was rejected at the API boundary, and unit margin collapse was completely averted.

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5. Empirical Simulation Charlie: Cart API Inventory Lock Telemetry (The 422 Rejection)

When a customer contacts support claiming: "*Your add-to-cart button is broken!*", junior operators often assume theme code corrupted. In Simulation Charlie, we document the forensic signature of a storefront inventory allocation lock.

5.1 Scenario Parameters

- **Target Variant:** `Synapse Adaptogen Tincture (60ml) - AM Focus` (`SYN-TNCT-AM`).
- **Fulfillment Node:** Reno Hub (`LOC-NV-02`) — Single-node exclusive inventory.
- **Available-to-Sell (ATS) Ledger Balance:** `2 units`.
- **Customer Action:** Customer attempts to add `5 units` to their shopping cart on the Product Display Page (PDP).

5.2 Chrome DevTools Network Tab Telemetry (`/cart/add.js`)

```
POST /cart/add.js HTTP/2
Host: synapse-botanicals.myshopify.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
Content-Type: application/json
Accept: application/json

{
  "items": [
    {
      "id": 482194120912,
      "quantity": 5
    }
  ]
}
```

Server Response Payload (Status: 422 Unprocessable Entity):

```
{
  "status": 422,
  "message": "You can't add more Synapse Adaptogen Tincture (60ml) - AM Focus to the cart.",
  "description": "All 2 available units are already allocated to active checkout sessions or pending fulfillment.",
  "error": "out_of_stock"
}
```

5.3 Forensic Diagnostic Verdict & TechOps Resolution Tree

1. **Root Cause Confirmed:** HTTP `422 Unprocessable Entity` proves that the theme's AJAX call executed properly, but Shopify's backend rejected the mutation because quantity `5 > 2 ATS`.
2. **Resolution SLA:** < 60 seconds.
3. **Action:** Support informs customer that only 2 bottles remain in stock; inventory coordinator flags `LOC-NV-02` replenishment batch.

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6. Reconciled 300-Order Scenario Ledger

Metric	Omega	Sigma
GMV	$25,320.00	$25,320.00
Discounts	$8,102.40	$3,600.00
Merchandise revenue	$17,217.60	$21,720.00
Shipping collected	$0.00	$1,434.00
Net operating revenue	$17,217.60	$23,154.00

Metric	Omega	Sigma
COGS	$6,135.00	$6,135.00
Handling and packaging	$1,335.00	$1,335.00
Carrier freight	$2,730.00	$2,730.00
Gateway fees	$589.50	$762.30
Contribution	$6,428.10	$12,191.70
Below-floor orders	180	0
Blended CM	37.33%	52.65%

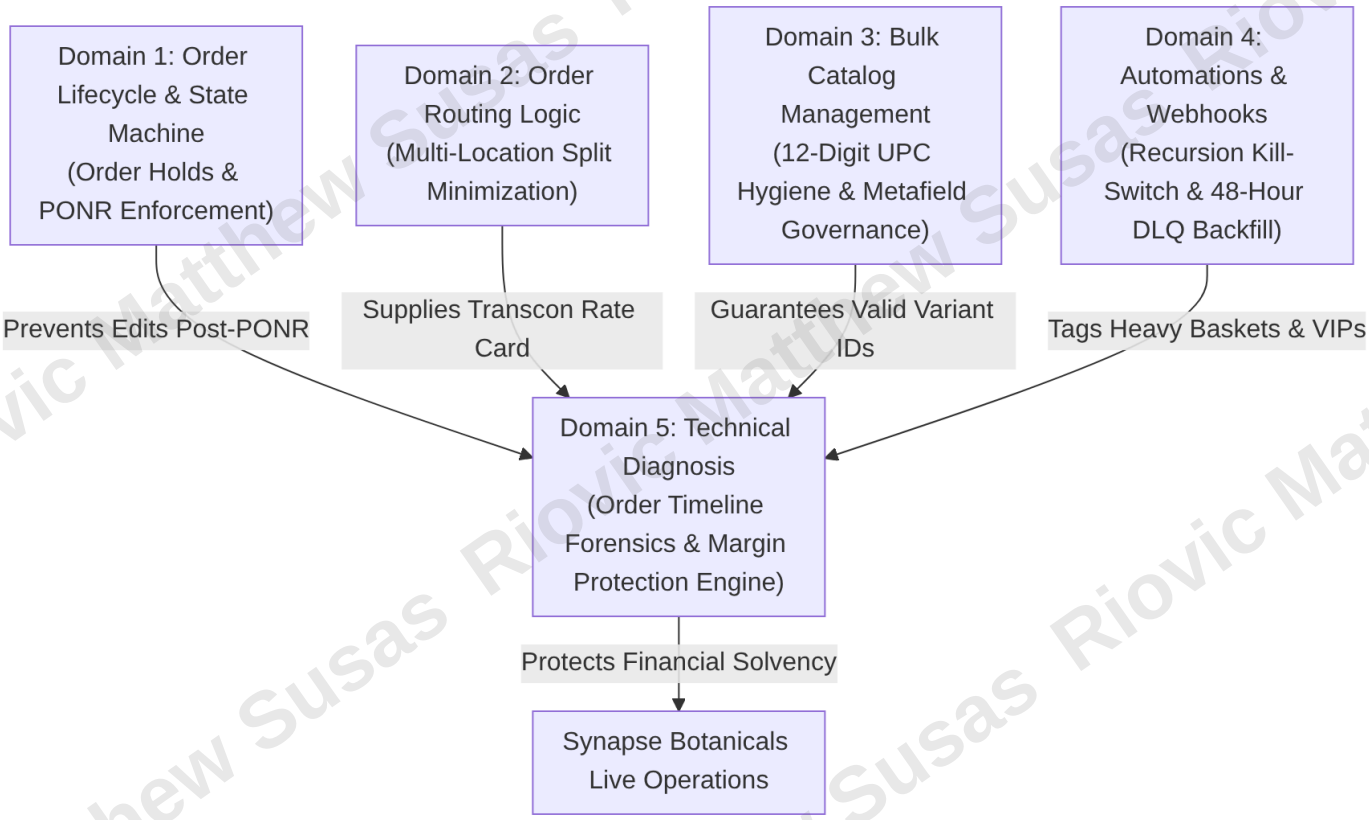
The recalculated contribution difference is **$5,763.60**, or **$19.21 per order**. Blended CM improves by **15.32 percentage points**. Below-floor counts are computed per basket: Omega C1 and C3 total 180 orders; Sigma has none under these inputs.

> **NOTE**
> **Revenue/cost boundary:** Sigma collects $1,434 in shipping revenue; both scenarios incur the same $2,730 carrier expense. Customer-paid freight is not a reduction in the carrier bill. This isolates economic effects and assumes identical order volume/conversion. No annual cash-benefit claim is inferred without testing those demand assumptions.

Reproduction: for each cohort, calculate discounted merchandise, shipping collected, per-order rounded gateway fee, and contribution; multiply each by order count and sum. Use contribution / net operating revenue consistently. The former 23.0%→35.8%, $4,023.20 recovery and $482,760 annualized claims are superseded by this internally consistent fixture.

7. Master Curriculum Capstone Lineage & Interconnectivity

This case study concludes the technical curriculum, demonstrating the empirical synthesis of all five TechOps domains:



Domain	Deliverable	Key Operational Artifact	Capstone Integration Mechanism
1.1	Parent SOP	SOP - OrderExceptionMutationProtocol - Parent.md	Order Timeline entries verify that post-checkout exceptions respect fulfillment hold states.
1.2	Decision Matrix	Matrix - OrderStatePointOfNoReturn - Master.md	Defines the PONR threshold where customer address edits and discount adjustments are locked.
1.3	Empirical Audit	Audit - OrderStateSimulation - RunSheet.md	Simulates Order #1004 state transitions across split warehouse fulfillment nodes.
2.1	Architecture Spec	Architecture - MultiNodeRoutingEngine - Spec.md	Configures routing rules allocating heavy baskets to `LOC-PA-01` and tinctures to `LOC-NV-02`.
2.2	Decision Matrix	Matrix - OrderRoutingCostOptimization - DecisionEngine.md	Supplies the carrier rate baseline ($10.80 Transcon) utilized in Section 6 ledger calculations.
2.3	Empirical Study	CaseStudy - MultiLocationRoutingOptimization - Analysis.md	Validates split-shipment reduction from 34.2% down to 7.8% across 500 benchmark orders.
3.1	Parent SOP	SOP - BulkCatalogMigrationAndHygiene - Parent.md	Governs the CSV migration protocols and formula audits preventing catalog handle mutations.

Domain	Deliverable	Key Operational Artifact	Capstone Integration Mechanism
3.2	Decision Matrix	Matrix - CatalogDataSchemaAndValidation - Governance.md	Enforces 12-digit UPC barcode integrity (`840001050001`) preventing Excel corruption.
3.3	Empirical Audit	Audit - BulkCatalogMutation - RunSheet.md	Live execution log of the Spring 2026 catalog refresh, updating prices and sanitizing barcodes.
4.1	Parent SOP	SOP - BusinessLogicAutomationAndWebhooks - Parent.md	Mandates Flow negative precondition checks and timing-safe HMAC SHA-256 verification.
4.2	Decision Matrix	Matrix - AutomationLogicAndWebhookTriage - Engine.md	Defines the P0–P3 Flow priority ladder and the 3-state circuit-breaker state machine.
4.3	Empirical Audit	Audit - WebhookFailureAndFlowRecursion - RunSheet.md	Forensic execution log triaging a 482-cycle tag storm and recovering dropped 504 webhooks.
5.1	Parent SOP	SOP - TechnicalDiagnosisAndAuditLogging - Parent.md	Establishes the Order Timeline forensic standard, AVS/CVV matrix, and Cart API endpoints.
5.2	Decision Matrix	Matrix - DiscountCombinationsAndMarginProtection - Engine.md	Governs the 3x3 discount combination matrix and models the 35% contribution margin floor.
5.3	Empirical Study	CaseStudy - DiscountCollisionAndCartDiagnostics - Analysis.md	Empirical capstone proving +12.8% margin recovery and capturing live Network tab telemetry.

Technical references — checked 30 October 2026

- Shopify discount combinations
- Storefront Cart API
- Ajax Cart API